Md Arif Shaikh

Department of Physics
Vivekananda Satavarshiki Mahavidyalaya
Manikpara, West Bengal, 721513, India

Email: arifshaikh.astro@gmail.com
ORCID: 0000-0003-0826-6164
Website: https://mdarifshaikh.com

Positions

Assistant Professor November 2023 – Present
Department of Physics, *Vivekananda Satavarshiki Mahavidyalaya*
Manikpara, West Bengal, 721513, India

Postdoctoral Fellow July 2022 – November 2023
Department of Physics and Astronomy, *Seoul National University*
1 Gwanak-ro, Gwanak-gu, Seoul 08826, Korea
Advisor: Hyung Mok Lee (hmlee@snu.ac.kr)

Postdoctoral Fellow July 2019 – June 2022
Astrophysical Relativity, *International Centre for Theoretical Sciences*
Hesaraghatta Hobli, Karnataka, 560089, India
Advisor: Parameswaran Ajith (ajith@icts.res.in)

Education

Doctor of Philosophy (PhD) July 2014 – June 2019
Harish-Chandra Research Institute
Chhatnag Road, Jhunsi, Prayagraj, 211019, Uttar Pradesh, India
Cosmology & High Energy Astrophysics
Advisor: Tapas Kumar Das (tapas@hri.res.in)

Master of Science (MSc) July 2012 – June 2014
Harish-Chandra Research Institute
Chhatnag Road, Jhunsi, Prayagraj, 211019, Uttar Pradesh, India
Cosmology & High Energy Astrophysics
Advisor: Tapas Kumar Das (tapas@hri.res.in)

Bachelor of Science (BSc) July 2009 – June 2012
Jadavpur University
188, Raja S.C. Mallick Rd, Kolkata 700032, India
Faculty of Physics

Awards, Fellowships & Memberships

- *ANRF International Travel Scheme (ITS) grant*, ANRF, Government of India., 2026
- *Core member*, LISA Consortium, 2025–
- *Offer for National Postdoctoral Fellowship (NPDF)*, SERB, Government of India., 2024
- *Membership*, Korean Astronomical Society, 2023–
- *Membership*, Simulating eXtreme Spacetime (SXS), 2021–
- *Membership*, LIGO Scientific Collaboration, 2020–
- *Senior Research Fellowship*, Department of Atomic Energy, Government of India, 2014–2018
- *Junior Research Fellowship*, Department of Atomic Energy, Government of India, 2012–2014
- *Offer for Phd in Physics starting from 2014*, IUCAA, pune, 2012
- *Offer for Integrated Phd in Physics*, NCRA-TIFR, Pune, 2012
- *Offer for Integrated Phd in Physics*, IISc, Bangalore, 2012

- *Offer for MSc in Physics*, IIT Bombay, 2012
- *AIR 41 in Joint Entrance Screening Test (JEST)*, JEST, 2012
- *AIR 43 in Joint Admission Test for M.Sc (JAM)*, IIT Bombay, 2012
- *INSPIRE Fellowship*, Department of Science and Technology, Government of India, 2009–2012
- *Ranked within top 20 in higher secondary examination*, West Bengal Council of Higher Secondary Education, 2009
- *Mamraj Agarwal Rashtriya Puraskar*, Governer of West Bengal, Shri Gopal Krishna Gandhi, 2007
- *First rank in secondary examination*, West Bengal Board of Secondary Education, 2007

Publications

Total publications: 76 (Short author publications: 22, SXS Collaboration: 1, LVK Collaboration: 53)

Short author publications

22. Ravichandran, A., Nee, P. J., Mitman, K., Islam, T., Field, S. E., Varma, V., Boyle, M., Ceja, A., Deppe, N., Ghadiri, N., Kidder, L. E., Kumar, P., Morales, M., Moxon, J., Nelli, K. C., Pfeiffer, H. P., Ramos-Buades, A., Rink, K., Rüter, H. R., Scheel, M. A., Shaikh, M. A., Stein, L. C., Tellez, D., Throwe, W., Vu, N. L. "Merger remnant and eccentricity dynamics surrogates for eccentric nonspinning black hole binaries.", 2026, arXiv:2605.00124

21. Islam, T., Ravichandran, A., Nee, P. J., Field, S. E., Varma, V., Pfeiffer, H. P., Ceja, A., Ghadiri, N., Kidder, L. E., Kumar, P., Morales, M., Ravishankar, A., Ramos-Buades, A., Rink, K., Ruter, H. R., Scheel, M. A., Shaikh, M. A., Tellez, D. "Including higher-order modes in a quadrupolar eccentric numerical relativity surrogate using universal eccentric modulation functions." *Phys. Rev. D*, **114**, *4*, 2026, arXiv:2604.17868

20. Maity, S., Tarafdar, P., Shaikh, M. A., Ghose, S., Das, T. K. "Effect of disc heights on acoustic surface gravity for accreting Kerr black holes." *Eur. Phys. J. C*, **86**, *2*, 2026

19. Nee, P. J., Ravichandran, A., Field, S. E., Islam, T., Pfeiffer, H. P., Varma, V., Boyle, M., Ceja, A., Ghadiri, N., Kidder, L. E., Kumar, P., Maurya, A., Morales, M., Ramos-Buades, A., Ravishankar, A., Rink, K., Rüter, H. R., Scheel, M. A., Shaikh, M. A., Tellez, D. "Eccentric binary black holes: A new framework for numerical relativity waveform surrogates." *Phys. Rev. Res.*, **8**, *2*, 2026, arXiv:2510.00106

18. Tiwari, A., Bhat, S. A., Shaikh, M. A., Kapadia, S. J. "Testing the nature of GW200105 by probing the frequency evolution of eccentricity." *Astrophys. J.*, **995**, *1*, 2025, arXiv:2509.26152

17. Bhat, S. A., Tiwari, A., Shaikh, M. A., Kapadia, S. J. "Eccentricity evolution consistency test to distinguish eccentric gravitational-wave signals from eccentricity mimickers." *Phys. Rev. D*, **112**, *12*, 2025, arXiv:2508.14850

16. Shaikh, M. A., Varma, V., Ramos-Buades, A., Pfeiffer, H. P., Boyle, M., Kidder, L. E., Scheel, M. A. "Defining eccentricity for spin-precessing binaries." *Class. Quant. Grav.*, **42**, 2025, arXiv:2507.08345

15. Chartier, N., Shaikh, M. A., Lee, H. M., Kim, J. "Comparison between best-fit eccentricity definitions and the standardized definition of eccentricity." *Phys. Rev. D*, **112**, *2*, 2025, arXiv:2503.19538

14. Deka, U., Prabhu, G., Shaikh, M. A., Kapadia, S. J., Varma, V., Field, S. E. "Surrogate modeling of gravitational waves microlensed by spherically symmetric potentials." *Phys. Rev. D*, **111**, *10*, 2025, arXiv:2501.02974

13. Deka, U., Chakraborty, S., Kapadia, S. J., Shaikh, M. A., Ajith, P. "Probing the charge of compact objects with gravitational microlensing of gravitational waves." *Phys. Rev. D*, **111**, *6*,

2025, arXiv:2401.06553

12. Shaikh, M. A., Bhat, S. A., Kapadia, S. J. "A study of the inspiral-merger-ringdown consistency test with gravitational-wave signals from compact binaries in eccentric orbits." *Phys. Rev. D*, **110**, *2*, 2024, arXiv:2402.15110

11. Shaikh, M. A., Varma, V., Pfeiffer, H. P., Ramos-Buades, A., van de Meent, M. "Defining eccentricity for gravitational wave astronomy." *Phys. Rev. D*, **108**, *10*, 2023, arXiv:2302.11257

10. Singh, M. K., Divyajyoti, D., Kapadia, S. J., Shaikh, M. A., Ajith, P. "Improved early-warning estimates of luminosity distance and orbital inclination of compact binary mergers using higher modes of gravitational radiation." *Mon. Not. Roy. Astron. Soc.*, **513**, *3*, 2022, arXiv:2202.05802

9. Maity, S., Shaikh, M. A., Tarafdar, P., Das, T. K. "Carter-Penrose diagrams for emergent spacetime in axisymmetrically accreting black hole systems." *Phys. Rev. D*, **106**, *4*, 2022, arXiv:2106.07598

8. Wei, W., Huerta, E. A., Yun, M., Loutrel, N., Shaikh, M. A., Kumar, P., Haas, R., Kindratenko, V. "Deep Learning with Quantized Neural Networks for Gravitational-wave Forecasting of Eccentric Compact Binary Coalescence." *Astrophys. J.*, **919**, *2*, 2021, arXiv:2012.03963

7. Singh, M. K., Kapadia, S. J., Shaikh, M. A., Chatterjee, D., Ajith, P. "Improved early warning of compact binary mergers using higher modes of gravitational radiation: A population study." *Mon. Not. Roy. Astron. Soc.*, **502**, *2*, 2021, arXiv:2010.12407

6. Kapadia, S. J., Singh, M. K., Shaikh, M. A., Chatterjee, D., Ajith, P. "Of Harbingers and Higher Modes: Improved gravitational-wave early-warning of compact binary mergers." *Astrophys. J. Lett.*, **898**, *2*, 2020, arXiv:2005.08830

5. Shaikh, M. A., Maity, S., Nag, S., Das, T. K. "Effective sound speed in relativistic accretion discs around Schwarzschild black holes." *New Astron.*, **69**, 2019, arXiv:1806.04084

4. Shaikh, M. A., Das, T. K. "Linear perturbations of low angular momentum accretion flow in the Kerr metric and the corresponding emergent gravity phenomena." *Phys. Rev. D*, **98**, *12*, 2018, arXiv:1803.09896

3. Shaikh, M. A.. "Relativistic sonic geometry for isothermal accretion in the Kerr metric." *Class. Quant. Grav.*, **35**, *5*, 2018, arXiv:1705.04918

2. Datta, S., Shaikh, M. A., Das, T. K. "Acoustic geometry obtained through the perturbation of the Bernoulli's constant." *New Astron.*, **63**, 2018, arXiv:1612.07954

1. Shaikh, M. A., Firdousi, I., Das, T. K. "Relativistic sonic geometry for isothermal accretion in the Schwarzschild metric." *Class. Quant. Grav.*, **34**, *15*, 2017, arXiv:1612.07963

SXS Collaboration

1. Scheel, M. A., others. "The SXS collaboration's third catalog of binary black hole simulations." *Class. Quant. Grav.*, **42**, *19*, 2025, arXiv:2505.13378

LVK Collaboration

53. Abac, N., others. "GWTC-5.0: An Introduction to Version 5.0 of the Gravitational-Wave Transient Catalog.", 2026, arXiv:2605.27223

52. Abac, A. G., others. "Searches for Binary Mergers with Sub-solar Mass Components in Data from the First Part of LIGO–Virgo–KAGRA's Fourth Observing Run.", 2026, arXiv:2605.05444

51. Abac, A. G., others. "GW240925 and GW250207: Astrophysical Calibration of Gravitational-wave Detectors.", 2026, arXiv:2605.11703

50. Abac, A. G., others. "Searches for Continuous Gravitational Waves from Supernova Remnants in the first part of the LIGO-Virgo-KAGRA Fourth Observing run.", 2026, arXiv:2603.25808

49. Abac, A. G., others. "Narrowband searches for continuous gravitational waves from known

pulsars in the first two parts of the fourth LIGO–Virgo–KAGRA observing run.", 2026, arXiv:2603.25938

48. Abac, A. G., others. "GWTC-4.0: Tests of General Relativity. III. Tests of the Remnants.", 2026, arXiv:2603.19021

47. Abac, A. G., others. "GWTC-4.0: Tests of General Relativity. I. Overview and General Tests.", 2026, arXiv:2603.19019

46. Abac, A. G., others. "GWTC-4.0: Tests of General Relativity. II. Parameterized Tests.", 2026, arXiv:2603.19020

45. Abac, A. G., others. "All-sky Searches for Continuous Gravitational Waves from Isolated Neutron Stars in the Data from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run.", 2026, arXiv:2603.14168

44. Abbasi, R., others. "Deep Search for Joint Sources of Gravitational Waves and High-energy Neutrinos with IceCube during the Third Observing Run of LIGO and Virgo." *Astrophys. J.*, **1003**, *1*, 2026, arXiv:2601.07595

43. Abac, A. G., others. "Constraints on gravitational waves from the 2024 Vela pulsar glitch.", 2025, arXiv:2512.17990

42. Abac, A. G., others. "GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures.", 2025, arXiv:2512.16347

41. Abac, A. G., others. "Search for planetary-mass ultra-compact binaries using data from the first part of the LIGO–Virgo–KAGRA fourth observing run.", 2025, arXiv:2511.19911

40. Abac, A. G., others. "All-sky search for continuous gravitational-wave signals from unknown neutron stars in binary systems in the first part of the fourth LIGO-Virgo-KAGRA observing run.", 2025, arXiv:2511.16863

39. Abac, A. G., others. "Direct multi-model dark-matter search with gravitational-wave interferometers using data from the first part of the fourth LIGO-Virgo-KAGRA observing run.", 2025, arXiv:2510.27022

38. Abac, A. G., others. "Cosmological and High Energy Physics implications from gravitational-wave background searches in LIGO-Virgo-KAGRA's O1-O4a runs.", 2025, arXiv:2510.26848

37. Abac, A. G., others. "GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-spin Black Hole Coalescences." *Astrophys. J. Lett.*, **993**, *1*, 2025, arXiv:2510.26931

36. Abac, A. G., others. "Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observing Run.", 2025, arXiv:2510.17487

35. Abac, A. G., others. "Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO-Virgo-KAGRA observing run.", 2025, arXiv:2509.07352

34. Abac, A. G., others. "GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes." *Phys. Rev. Lett.*, **135**, *11*, 2025, arXiv:2509.08054

33. Abac, A. G., others. "Black Hole Spectroscopy and Tests of General Relativity with GW250114." *Phys. Rev. Lett.*, **136**, *4*, 2026, arXiv:2509.08099

32. Abac, A. G., others. "GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation.", 2025, arXiv:2509.04348

31. Abac, A. G., others. "Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run.", 2025, arXiv:2508.20721

30. Abac, A. G., others. "Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run.", 2025, arXiv:2508.18079

29. Abac, A. G., others. "GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run.", 2025, arXiv:2508.18082

28. Abac, A. G., others. "GWTC-4.0: Population Properties of Merging Compact Binaries.", 2025, arXiv:2508.18083

27. Abac, A. G., others. "GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients.", 2025, arXiv:2508.18081

26. Abac, A. G., others. "GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog." *Astrophys. J. Lett.*, **995**, *1*, 2025, arXiv:2508.18080

25. Abac, A. G., others. "All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA observing run." *Phys. Rev. D*, **113**, *8*, 2026, arXiv:2507.12282

24. Abac, A. G., others. "All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO-Virgo-KAGRA observing run." *Phys. Rev. D*, **112**, *10*, 2025, arXiv:2507.12374

23. Abac, A. G., others. "GW231123: A Binary Black Hole Merger with Total Mass 190–265 $M_\odot$." *Astrophys. J. Lett.*, **993**, *1*, 2025, arXiv:2507.08219

22. Scheel, M. A., others. "The SXS collaboration's third catalog of binary black hole simulations." *Class. Quant. Grav.*, **42**, *19*, 2025, arXiv:2505.13378

21. Abac, A. G., others. "Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run." *Astrophys. J.*, **983**, *2*, 2025, arXiv:2501.01495

20. Abac, A. G., others. "Search for Gravitational Waves Emitted from SN 2023ixf." *Astrophys. J.*, **985**, *2*, 2025, arXiv:2410.16565

19. Abac, A. G., others. "A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154." *Astrophys. J.*, **977**, *2*, 2024, arXiv:2410.09151

18. Raman, G., others. "Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run." *Astrophys. J.*, **980**, *2*, 2025, arXiv:2407.12867

17. Abac, A. G., others. "Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_\odot$ Compact Object and a Neutron Star." *Astrophys. J. Lett.*, **970**, *2*, 2024, arXiv:2404.04248

16. Abac, A. G., others. "Ultralight vector dark matter search using data from the KAGRA O3GK run." *Phys. Rev. D*, **110**, *4*, 2024, arXiv:2403.03004

15. Fletcher, C., others. "A Joint Fermi-GBM and Swift-BAT Analysis of Gravitational-wave Candidates from the Third Gravitational-wave Observing Run." *Astrophys. J.*, **964**, *2*, 2024, arXiv:2308.13666

14. Abac, A. G., others. "Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo." *Astrophys. J.*, **973**, *2*, 2024, arXiv:2308.03822

13. Abbott, R., others. "Search for Gravitational-lensing Signatures in the Full Third Observing Run of the LIGO–Virgo Network." *Astrophys. J.*, **970**, *2*, 2024, arXiv:2304.08393

12. Abbott, R., others. "Open Data from the Third Observing Run of LIGO, Virgo, KAGRA, and GEO." *Astrophys. J. Suppl.*, **267**, *2*, 2023, arXiv:2302.03676

11. Abbott, R., others. "Search for subsolar-mass black hole binaries in the second part of Advanced LIGO's and Advanced Virgo's third observing run." *Mon. Not. Roy. Astron. Soc.*, **524**, *4*, 2023, arXiv:2212.01477

10. Abbott, R., others. "Search for Gravitational-wave Transients Associated with Magnetar Bursts in Advanced LIGO and Advanced Virgo Data from the Third Observing Run." *Astro-*

phys. J., **966**, *1*, 2024, arXiv:2210.10931

9. Abbott, R., others. "Model-based Cross-correlation Search for Gravitational Waves from the Low-mass X-Ray Binary Scorpius X-1 in LIGO O3 Data." *Astrophys. J. Lett.*, **941**, *2*, 2022, arXiv:2209.02863
8. Abbott, R., others. "Search for continuous gravitational wave emission from the Milky Way center in O3 LIGO-Virgo data." *Phys. Rev. D*, **106**, *4*, 2022, arXiv:2204.04523
7. Abbott, R., others. "First joint observation by the underground gravitational-wave detector KAGRA with GEO 600." *PTEP*, **2022**, *6*, 2022, arXiv:2203.01270
6. Abbott, R., others. "Search for gravitational waves from Scorpius X-1 with a hidden Markov model in O3 LIGO data." *Phys. Rev. D*, **106**, *6*, 2022, arXiv:2201.10104
5. Abbott, R., others. "All-sky search for continuous gravitational waves from isolated neutron stars using Advanced LIGO and Advanced Virgo O3 data." *Phys. Rev. D*, **106**, *10*, 2022, arXiv:2201.00697
4. Abbott, R., others. "Tests of General Relativity with GWTC-3." *Phys. Rev. D*, **112**, *8*, 2025, arXiv:2112.06861
3. Abbott, R., others. "All-sky search for gravitational wave emission from scalar boson clouds around spinning black holes in LIGO O3 data." *Phys. Rev. D*, **105**, *10*, 2022, arXiv:2111.15507
2. Abbott, R., others. "Searches for Gravitational Waves from Known Pulsars at Two Harmonics in the Second and Third LIGO-Virgo Observing Runs." *Astrophys. J.*, **935**, *1*, 2022, arXiv:2111.13106
1. Abbott, R., others. "Constraints on the Cosmic Expansion History from GWTC–3." *Astrophys. J.*, **949**, *2*, 2023, arXiv:2111.03604

Seminars

14. "Defining eccentricity for spin-precessing binaries", ICTS, Bangalore, India, August 7, 2025
13. "Defining eccentricity for spin-precessing binaries", IUCAA, Pune, India, June 18, 2025
12. "gw_eccentricity: a Python package to measure orbital eccentricity from gravitational waveforms", Inje University, Gimhae, Korea, November 3, 2023
11. "gw_eccentricity: a Python package to measure orbital eccentricity from gravitational waveforms", IUCAA, Pune, India, June 19, 2023
10. "gw_eccentricity: a Python package to measure orbital eccentricity from gravitational waveforms", ICTS-TIFR, Bangalore, India, June 15, 2023
9. "Measuring Eccentricity from Gravitational Waveform", ITP, Beijing, China, May 11, 2023
8. "Measuring Eccentricity from Gravitational Waveform", BNU, Beijing, China, May 10, 2023
7. "Defining eccentricity for gravitational wave astronomy", KASI, Daejeon, Korea, April 20, 2023
6. "Standardizing the definition of eccentricity for gravitational wave astronomy", IBS, Daejeon, Korea, April 18, 2023
5. "Defining eccentricity for gravitational wave astronomy", ICTS-TIFR, Bangalore (online), India, December 1, 2022
4. "Defining eccentricity for gravitational wave astronomy", RESCEU, Tokyo, Japan, November 18, 2022
3. "Defining eccentricity for gravitational wave astronomy", OIT, Osaka, Japan, November 16, 2022

2. "Defining eccentricity for gravitational wave astronomy", YITP, Kyoto, Japan, November 14, 2022

1. "On the emergent sonic geometry through the linear perturbation of relativistic black hole accretion", HRI, Allahabad, India, November 18, 2019

Conference Talks

9. "What do we know about the eccentricity of the GW sources?", Cosmology in Multi-Messenger Era, Seoul, Korea, October 14–14, 2024

8. "A study of IMRCT on eccentric GW signal", GW Universe Workshop 8, Seoul, Korea, December 26–27, 2023

7. "Comparing eccentric waveforms for gravitational wave data analysis", 2nd H.S. Yun Astronomy Workshop, Seoul, Korea, August 29–30, 2023

6. "Defining eccentricity for gravitational wave astronomy", ICGAC15, Gyeongju, Korea, July 3–7, 2023

5. "Defining eccentricity for gravitational wave astronomy", APSWGC, Hangzhou, China, May 14–22, 2023

4. "Defining eccentricity for gravitational wave astronomy", GWPAW, Melbourne, Australia, December 4–9, 2022

3. "Probing the evolution history of compact binaries from higher modes of gravitation waves", ICTS In-house symposium, Bangalore, India, February 17–18, 2020

2. "Relativistic acoustic geometry in general relativistic accretion disc around Kerr black holes", Exploring the Universe: Near Earth Space Science to Extra-Galactic Astronomy, Kolkata, India, November 14–17, 2018

1. "Emergence of curved sonic manifold for isothermal accretion in black hole metric", Young Astronomers Meet, Pune, India, September 11–15, 2017

Conference Posters

5. "Surrogate and hybrid models of eccentric waveforms using numerical relativity", 765. WE-Heraeus-Seminar: Gravitational Wave and Multimessenger Astronomy, Physikzentrum Bad Honnef, Germany, April 25–28, 2022

4. "Probing the evolution of compact binaries using higher modes of gravitational waves", 39th meeting of ASI, Online, India, February 18–23, 2021

3. "Probing evolution history of compact binaries using higher modes of gravitational waves", LVK September Meeting, Online, LIGO, September 14–17, 2020

2. "Relativistic sonic geometry for isothermal accretion in Kerr metric", 29th meeting of IAGRG, Guwahati, India, May 18–20, 2017

1. "Emergence of relativistic sonic geometry through perturbation of matter in black hole metric", 35th meeting of ASI, Jaipur, India, March 6–10, 2017

Workshops

26. "LIGO-Virgo-KAGRA Collaboration Meeting", IUCAA, Pune, India, September 21–25, 2026

25. "The Future of Gravitational-Wave Astronomy", ICTS, Bangalore, India, October 27–31, 2025

24. "Cosmology in Multi-Messenger Era", Gravitational Wave Universe, SNU, Seoul, Korea, October 14–14, 2024

23. "KAS 2023 Fall Meeting", Korean Astronomical Society, RABADA PLAZA, JEJU, Korea, Oc-

tober 18–20, 2023

22. "LVK 2023 September Meeting", LVK Collaboration, Toyama, Japan, September 11–15, 2023
21. "GW Universe Workshop 7", Gravitational Wave Universe, Seoul, Korea, June 12–12, 2023
20. "KAS 2023 Spring Meeting", Korean Astronomical Society, LAHANHOTEL, JEONJU, Korea, April 12–14, 2023
19. "LVK 2023 March Meeting", Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA), (remote participation), Evanston, USA, March 13–16, 2023
18. "68th Workshop on Gravitational Waves and Numerical Relativity", APCTP, Pohang, Korea, March 15–16, 2023
17. "GW Universe Winter Workshop 2023", Gravitational Wave Universe, Yongpyong Ski Resort, Gwangwon, Korea, February 26–March 1, 2023
16. "GW Universe Workshop6", Gravitational Wave Universe, Seoul, Korea, November 24–25, 2022
15. "67th Workshop on Gravitational Waves and Numerical Relativity", APCTP, Pohang, Korea, October 26–27, 2022
14. "Summer School on Gravitational-Wave Astronomy", ICTS-TIFR, Bangalore, India, July 5–16, 2021
13. "ICERM Fall 2020 Workshop 1: Advances and Challenges in Computational Relativity", ICERM, Brown University, Online, USA, September 14–18, 2020
12. "ICTS workshop on parameter estimation with bilby", ICTS-TIFR, Bangalore, India, August 27–28, 2020
11. "Test of General Relativity using Gravitational Waves", IACS, Kolkata \{}& IIT Gandhinagar, Online, India, August 13–14, 2020
10. "PyCBC Inference Online Workshop 2020", AEI, Hannover, Online, Germany, June 25–26, 2020
9. "Gravitational-Wave Open Data Workshop #3", LIGO-Virgo Collaboration, Online, USA, May 26–28, 2020
8. "Summer School on Gravitational-Wave Astronomy", ICTS-TIFR, Online, India, May 18–23, 2020
7. "Astrophysics of Supermassive Black Holes", ICTS-TIFR, Bangalore, India, December 17–19, 2019
6. "Newton Bhabha-Open Data workshop", IUCAA, Pune, India, December 4–6, 2019
5. "The Future of Gravitational-Wave Astronomy", ICTS, Bangalore, India, August 19–22, 2019
4. "Summer School on Gravitational-Wave Astronomy", ICTS-TIFR, Bangalore, India, July 25–26, 2019
3. "Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation", Galileo Galilei Institute for Theoretical Physics, Florence, Italy, March 11–22, 2019
2. "Black Holes: From Classical to Quantum Gravity", IIT, Gandhinagar, India, December 15–19, 2017
1. "Summer School on Gravitational-Wave Astronomy", ICTS-TIFR, Bangalore, India, July 17–28, 2017

Research Visits

20. Chunglee Kim, Ewha Womans University, Seoul, Korea, October 20–21, 2025

19. Hyung Mok Lee, Seoul National University, Seoul, Korea, October 15–25, 2025
18. Prayush Kumar, ICTS-TIFR, Bangalore, India, August 4–10, 2025
17. Shasvath Kapadia, IUCAA, Pune, India, June 16–22, 2025
16. Hyung Mok Lee, Seoul National University, Seoul, Korea, October 10–20, 2024
15. Hyung Won Lee, Inje University, Gimhae, Korea, November 2–3, 2023
14. Shasvath Kapadia, IUCAA, Pune, India, June 18–21, 2023
13. Prayush Kumar, ICTS-TIFR, Bangalore, India, June 15–18, 2023
12. Zhoujian Cao, BNU, Beijing, China, May 09–11, 2023
11. Arman Shafieloo, KASI, Daejeon, Korea, April 19–20, 2023
10. Young Bok Bae, IBS, Daejeon, Korea, April 18–19, 2023
9. Junichi Yokoyama, UoT, Tokyo, Japan, November 17–19, 2022
8. Hisa-aki Shinkai, OIT, Osaka, Japan, November 15–17, 2022
7. Kunihito Ioka, YITP, Kyoto, Japan, November 13–15, 2022
6. Frank Ohme, AEI, Hannover, Germany, April 20–24, 2022
5. Harald Pfeiffer, AEI, Potsdam, Germany, March 20–April 20, 2022
4. Tapas Kumar Das, HRI, Allahabad, India, November 15–25, 2019
3. P Ajith, ICTS-TIFR, Bangalore, India, March 25–April 6, 2019
2. P Ajith, ICTS-TIFR, Bangalore, India, September 27–October 11, 2017
1. Tarun Souradeep, IUCAA, Pune, India, September 6–October 20, 2014

Teaching

- **Stochastic gravitational wave background from early universe** (Tutor), Instructor: Shi Pi
  Summer School on Gravitational Wave Astronomy, ICTS-TIFR, Bangalore, India
  July 5–16, 2021
- **Introduction to General Relativity** (Tutor), Instructor: Bala Iyer
  Graduate semester course, ICTS-TIFR, Bangalore, India
  August 1–December 31, 2020
- **Parameter estimation with bilby** (Tutor), Instructor: Gregory Ashton
  ICTS workshop, ICTS-TIFR, Bangalore, India
  August 27–28, 2020
- **Numerical Hydrodynamics** (Tutor), Instructor: Ian Hawke
  Summer School on Gravitational Wave Astronomy, ICTS-TIFR, Bangalore, India
  May 18–June 5, 2020
- **An Introduction to GW Physics & Astronomy** (Tutor), Instructor: P. Ajith & Bala Iyer
  Graduate semester course, ICTS-TIFR, Bangalore, India
  January 1–April 30, 2020
- **Advanced General Relativity** (Tutor), Instructor: Sudipta Sarkar
  Summer School on Gravitational Wave Astronomy, ICTS-TIFR, Bangalore, India
  July 15–26, 2019
- **Statistical Physics** (Tutor), Instructor: G. V. Pai
  Graduate semester course, HRI, Allahabad, India
  August 1–December 31, 2017

Refereeing

- **Classical and Quantum Gravity**
- **The European Physical Journal C**
- **Journal of Physics A: Mathematical and Theoretical**

References

Tapas Kumar Das (Phd Advisor)
Professor, Harish-Chandra Research Institute
Chhatnag Road, Jhunsi, Prayagraj, 211019, Uttar Pradesh, India
Email: tapas@hri.res.in, Phone: +91 5322274321, Website: http://www.hri.res.in/ tapas/

Parameswaran Ajith (Postdoc mentor)
Professor, International Centre for Theoretical Sciences
Hesaraghatta Hobli, Karnataka, 560089, India
Email: ajith@icts.res.in, Phone: +91 8046536210, Website: https://home.icts.res.in/ ajith/Home.html

Hyung Mok Lee (Postdoc mentor)
Professor, Seoul National University
1 Gwanak-ro, Gwanak-gu, Seoul 08826, Korea
Email: hmlee@snu.ac.kr, Phone: 82-2-880-6625, Website: http://astro.snu.ac.kr/ hmlee/

Shasvath Kapadia
Assistant Professor, Inter-University Centre for Astronomy and Astrophysics
Post Bag 4, Ganeshkhind, Pune, Maharashtra 411007, India
Email: shasvath.kapadia@iucaa.in, shasvath.j.kapadia@gmail.com

Vijay Varma
Assistant Professor, University of Massachusetts
Dartmouth, USA
Email: vijay.varma392@gmail.com, Website: https://vijayvarma.com/

Prayush Kumar
Reader, International Centre for Theoretical Sciences
Hesaraghatta Hobli, Karnataka, 560089, India
Email: prayush@icts.res.in, Phone: +91 8046536390